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**Roll No.** | | | | | | | | | | | | | **Total No. of Pages : 02**

**Total No. of Questions : 18**

**B.Tech. (CE) (2012 to 2017) (Sem.–6)**

# FOUNDATION ENGINEERING

**Subject Code : BTCE-603**

**M.Code : 71084**

**Time : 3 Hrs.** **Max. Marks : 60**

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

## SECTION-A

**Answer briefly :**

1. What do you understand by site investigation?
2. What is reconnaissance? What is its use?
3. Define Earth Pressure at rest.
4. Define the following terms :
   a) Net safe bearing capacity.
   b) Gross safe bearing capacity
5. Write short note on Vesic's bearing capacity theory.
6. Differentiate between the general shear failure and the local shear failure.
7. Discuss the effect of water table on the bearing capacity of the soil.
8. How would you fix the depth of foundation? Mention the formula used generally.
9. Enumerate various types of pile foundations.
10. Write whether the following statements are true or false with justification:
    a) A disturbed sample is not a truly representative sample.
    b) A disturbed sample is absolutely undisturbed.

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## SECTION-B

11. What is the basic difference between a drilled pier and a caisson? What are the conditions in which a drilled pier is more suitable than a caisson?

12. How would you estimate the load carrying capacity of drilled pier in sand?

13. Describe any two methods for the construction of drilled piers.

14. What are different types of earth pressure? Give examples.

15. What are the assumptions in Coulomb's theory?

## SECTION-C

16. Determine the ultimate bearing capacity of a strip footing, 1.20 m wide, and having the depth of foundation of 1.0 m. Use Terzaghi's theory and assume general shear failure. Take $\Phi' = 35°$, $\gamma = 18$ kN/m$^3$ and $c' = 15$ kN/m$^2$.

17. Describe the procedure for the design of a strap footing.

18. How would you fix the depth of foundation? Discuss Rankine's formula for the minimum depth.

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